

Test Plan 3 — Product Matrix & Role Data Contracts (Offline)

Project	product
Specs	tests/product/matrix/product-matrix-data-contract.spec.ts , tests/product/matrix/product-roles-contract.spec.ts
Target	Offline — assertions over checked-in fixtures behind the e2e-matrix.data barrel
Cases	23 (13 matrix + 10 roles)
Skips	None — network-free, always runs
Skill	product-matrix-contract

Scope

Data-contract guards that run with no network access. "Data contract" means the tests assert the shape and rules of checked-in fixture data rather than any live behavior. They cover two things: the property-characterization **matrix** (a grid of scenarios × personas) and the three sign-in **roles**. The tests encode the rules from the source document, so any drift in the fixtures fails fast. They also keep runtime-only values — live tokens, computed prices — out of the committed data. No browser and no dev app are involved.

Preconditions

- None beyond a typecheck. The data lives in the e2e-matrix.data files, and no secrets are required.

Cases

product-matrix-data-contract.spec.ts — "Product matrix data contract" (@product)

ID	CASE	ASSERTS
MTX-01	Keeps runtime-only values out of checked-in matrix data	No live/computed runtime values committed

ID	CASE	ASSERTS
MTX-02	Keeps the scenario × persona matrix at its expected size	Matrix dimensions match the documented size
MTX-03	Matches the required CALC-ROOF main scenario ids from the document	All required CALC-ROOF ids present
MTX-04	Covers every calculator property type from the document	Every property type represented
MTX-05	Covers building, parking, and sports-court polygon behavior	Each polygon behavior covered
MTX-06	Covers all roof and surface types from the document	Every roof/surface type present
MTX-07	Keeps roof payload placeholders empty in checked-in data	Roof payload placeholders remain empty pre-run
MTX-08	Marks no-panel scenario as UI-only and keeps it out of API/JSON E2E flows	No-panel scenario flagged UI-only, excluded from API flows
MTX-09	Assigns a unique id to every matrix scenario	Scenario ids are unique
MTX-10	Keeps arena type consistent with each scenario group	Arena type matches scenario group
MTX-11	Keeps the negative scenario below the quotable panel minimum	Negative scenario sits under the quotable minimum
MTX-12	Grants elevated company privileges to exactly one persona	Exactly one persona has elevated company rights
MTX-13	Aligns panel counts with each scenario panel mode	Panel counts consistent with panel mode

product-roles-contract.spec.ts — "Product role contract" (@product)

ID	CASE	ASSERTS
ROL-01	Models all three sign-in roles as distinct personas	customer / consultant / company are distinct
ROL-02	Every sign-in role has a human-readable name and role description	Name + description present per role

ID	CASE	ASSERTS
ROL-03	Post-funding destination is a known value for every sign-in role	Destination is an allowed enum value
ROL-04	Customer is the only sign-in role that lands on quotations after funding	Only customer → quotations post-funding
ROL-05	Customer can open quotations from the results screen	Customer has quotations access from results
ROL-06	Consultant reaches results but can still open quotations	Consultant result flow + quotations access
ROL-07	Company downloads its own quotation from results, not the quotations list	Company downloads own quotation from results
ROL-08	Only company can open company pricing	Company-only pricing access
ROL-09	Only company can open company management	Company-only management access
ROL-10	Company is the only role with any elevated company rights	Elevated rights are exclusive to company

Run

```
npx playwright test --project=product tests/product/matrix
```

Notes

- This group is the offline counterpart to Plan 4 (live role e2e). The role contract here asserts the *rules*; Plan 4 confirms that a live session actually *behaves* by those rules.
- When the source document changes, edit the data in the `e2e-matrix.data` files, not the specs. The contract then re-validates the new data.